SARTHAK PAINULI

Software Engineer

+91 9318461676 | sarthak.painuli3101@gmail.com | Bengaluru, Karnataka | github.com/sarthak310104 | linkedin.com/in/sarthak-painuli

SUMMARY

Software Engineer with ~1 year of professional experience and 1+ year of internship experience. I've worked across the full stack: APIs, payment flows, databases, and the frontends that sit on top of them. I enjoy digging into why something breaks under real conditions, understanding the problem properly, and building something that actually holds up under load.

SKILLS

Languages: Java, Python, TypeScript, Haskell

Frameworks: FastAPI, Node.js, Django, Flask

Backend & APIs: REST APIs, JWT Authentication, SQL, PostgreSQL, MongoDB, Redis, Task Queues (Celery), Alembic, OpenTelemetry (OTLP)

Computer Science Fundamentals: Data Structures & Algorithms, Object-Oriented Programming

DevOps & Tools: Git, Linux, Docker, Docker Compose, Nginx, Postman, AWS, GitHub Actions (CI/CD)

EXPERIENCE

Product Engineer - Juspay 08/2025 – 06/2026

Bengaluru, India

- Built Google Wallet and Apple Pay support into a merchant checkout screen as part of a multi-payment-method redesign, making the implementation config-driven across merchants.
- Owned 6+ transaction flows end to end, including initiation, 3DS authentication, refunds, captures, and webhook ingestion, across 3+ payment providers, building APIs and business logic for payment processing.
- Worked with the platform team to optimize backend modules in the payment flow's critical path, reducing redundant processing and improving load times and overall application performance by 15%.
- Built an MCP-based personalization tool that uses an LLM to fetch a prospective merchant's brand colors, logo, and currency and apply them to demo checkout configurations.

Web Development Intern - NXP Semiconductors 07/2024 – 06/2025

Noida, India

- Built 8+ internal tools using Java, JSP, and SQL, replacing manual workflows across multiple teams and working directly with stakeholders to define requirements.
- Designed a lab inventory and booking system for tracking and reserving 100K+ tools in real time, adding database indexes for fast availability lookups and logic to prevent double-booking.
- Reworked core workflows on an internal dev tooling platform, fixing reliability issues that had been blocking adoption, and grew usage by 40%.

PROJECTS

Ghost Protocol **github.com/sarthak310104/Ghost-Protocol-emp** Live Demo: ghost-protocol-emp.vercel.app

Python, FastAPI, Next.js/TypeScript, PostgreSQL, Redis, Celery

- Built a FastAPI/AsyncIO backend ingesting OTLP traces and metrics, auto-deriving a behavioral graph across services to flag anomalies and structural bottlenecks before they cause incidents.
- Built two independent incident-evidence engines: a mean-reversion simulator projecting recovery with 95% confidence intervals, and a concurrent-cohort comparator analyzing live canary/rollout traffic splits.
- Load-tested the ingestion pipeline with a custom benchmark, surfaced and fixed a Postgres deadlock and a write-race condition, sustaining 11k–23k spans/sec with API p95 latency under 14ms.
- Shipped an 11-page Next.js dashboard with session auth and workspace isolation, backed by full CI (78 automated tests) and a zero-cost live deployment.

Control-Tower **github.com/sarthak310104/Control-Tower**

Python, FastAPI, PostgreSQL, APScheduler, MCP, Docker

- Built a coordination service that lets multiple agents edit the same codebase without locking entire files.
- Used AST parsing to lock individual functions and kept claims valid when the surrounding code changed.
- Added automatic cleanup for expired claims and a live view of which agent owns what code.
- Added a messaging and activity-feed layer so agents can leave context and updates for each other.

EDUCATION

B.Tech in Computer Science Engineering 11/2021 – 06/2025

Jaypee Institute of Information Technology

ACHIEVEMENTS

- 3-star CodeChef, 1722 peak rating
- 150+ LeetCode problems solved

SOCIALS

LeetCode: leetcode.com/u/sarthak310104 | CodeChef: codechef.com/users/galahad_3101

HackerRank: hackerrank.com/profile/sarthakpainuli11 | Portfolio: sarthak-painuliportfolio.vercel.app